



IILM University, Greater Noida

Subject Name: Semiconductor and Quantum Physics

Course Code: 26SOST101

ASSIGNMENT

UNIT 1

Short Question

1. Briefly explain metals, semiconductors and insulators based on band gap.
2. With the help of diagram define valence band, conduction band and fermi level in solids.
3. Differentiate between n-type and p-type extrinsic semiconductors.
4. Write the concept of holes. When are the effective mass of holes assumed same as that of electron?
5. Schematically show how $F(E)$, Fermi function, varies with energy for any metal at following temperatures:
 - (i) at $T = 0\text{ K}$
 - (ii) at $T = \text{finite}$
6. Discuss the characteristics of Superconductors in superconducting state.
7. Write the short note on Meissner Effect.

Long Question

8. With the help of energy diagram briefly explain how bands are formed in solids. Differentiate semiconductors, conductor and insulators on the basis of band theory of solids.
9. What is Fermi-Dirac Distribution function? Give its physical significance. Schematically show that it varies with energy at different temperatures including absolute zero of temperature and finite temperature.

10. What is superconductivity? Discuss the temperature dependence of resistivity in superconducting materials. Also, discuss the significance of critical temperature and critical magnetic field for superconductors?
11. Deduce the expression for the carrier concentration in intrinsic semiconductors. Prove that Fermi level lies in the middle of the forbidden band in case of intrinsic semiconductors.

UNIT 2

Short Question

1. Differentiate between direct and indirect bandgap.
2. Write the expression of efficiency of photovoltaic cell.
3. Explain the concept of biasing in a p-n junction diode in brief, describing its two types..
4. Briefly describe the depletion region.
5. Discuss photovoltaic effect.

Long Question

6. What is a pn junction diode? Explain the formation of barrier potential and depletion region in a p-n junction diode.
7. Explain the flow of charge carriers in forward and reverse bias condition and hence draw the I-V characteristics of p-n junction diode.
8. What is a photovoltaic cell? Explain the working of PV cell with suitable diagram.
9. Compare direct and indirect bandgap semiconductors using suitable energy-band diagrams.

UNIT 3

Short Question

1. State de-Broglie's hypothesis of matter waves and determine the De Broglie wavelength of a particle.
2. State inadequacies of classical physics.
3. Define the wave function ψ in quantum mechanics and state its physical significance.
4. Briefly discuss the normalization condition for the wave function.
5. State the quantum tunneling effect.
6. Briefly explain quantum computing.

Long Question

7. Derive the time-dependent Schrödinger wave equation.
8. Derive the Schrödinger time-independent wave equation. Explain the physical significance of the Schrödinger wave function.
9. Apply the time-independent Schrödinger equation to a particle confined in a one-dimensional box. Obtain expressions for the energy eigenvalues and the normalized wave function.
10. Compare classical and quantum predictions for a particle encountering a potential barrier to explain the tunneling effect.
11. What are matter waves? Show that the wavelength λ associated with a particle of mass m and kinetic energy E is given by $\lambda = \frac{h}{\sqrt{2mE}}$.

UNIT 4

Short Question

1. Discuss the necessary conditions for lasing action.
2. Identify the basic components of a laser and explain the function of each.
3. Explain the importance of an optical resonator in the functioning of a laser.
4. What is the working principle of an optical fibre? Name and define it.
5. Briefly describe the working principle of semiconductor laser.

Long Question

1. Derive the relations between Einstein's A and B coefficients using the concept of thermal equilibrium and Planck's radiation law.
2. Explain the construction and working of a He–Ne laser with a neat labelled diagram. Also describe its energy level scheme and applications.
3. Explain how a Ruby laser is constructed and how it works. Illustrate with a diagram and describe its energy levels.
4. Discuss the classification of optical fibres. Explain step-index and graded-index fibres, and distinguish between single-mode and multimode fibres.
5. How do acceptance angle and numerical aperture control the entry and propagation of light in an optical fibre? Derive their expressions.